Capstone Project Phase 1 Report

AI251P Applied Computer Vision | Smart Pet Vision | M1–M3 | 40 points

We compare a custom CNN, a frozen ResNet-50 and a fully fine-tuned ResNet-50 for cat and dog classification. The selected fine-tuned model achieves 99.9096% test accuracy and 0.998965 macro-F1 on 1,106 held-out images.

1 Team and project

Field	Details
Team	Smart Pet Vision
Topic	Pet photo review for shelter adoption listings (original topic).
Application	Animal welfare and preparation of adoption-listing photographs
Repository	https://github.com/Yasser-20-alhajlah/smart-pet-vision
Published source snapshot	edcc29f2e035b329e577cd04bf6092dc1f4ae04c (29 October 2026). This snapshot does not identify the historical training commit.
Experiments	https://wandb.ai/yasser-alhajlah-kaust/smart-pet-vision
Dataset card	Appendix A and smart-pet-vision/docs/dataset_card.md

Team members

Member	ID	Phase 1 contribution
Abdulrahman Mastour Alharthi	(ID)	Data cleaning, split and audits; classifiers and evaluation; report and evidence
Yasser Mansour Alhajlah	(ID)	Derived body boxes and box review; W&B; clean reproduction recorded on 23 October
Mohammed Adeeb Alrashdi	(ID)	Trimap masks and team-photo annotation guide
Musaad Obaied Saad Alqahtani	(ID)	Repository, configuration, W&B setup and reproduction instructions

Each of the four team members reviewed and corrected approximately 100 photographs in CVAT.

Problem and acceptance criteria

Summary: The proposed system supports staff review of adoption-listing photographs. Phase 1 delivers the data foundations and whole-image cat/dog classifier. Body detection, segmentation, cut-outs and workflow integration belong to Phase 2.

Intended users: animal-shelter and adoption-platform staff. Staff retain publishing authority. Low confidence, classifier/detector disagreement or no detected animal should trigger referral. The 0.90 confidence threshold is provisional until evaluated on an independent calibration set.

Our Phase 1 targets are test accuracy ≥ 0.90 and test macro-F1 ≥ 0.90 for the classifier selected using validation data. These are project targets.

2 Data foundations M1

2.1 Datasets

Dataset	Contents	Use	Licence or access
Oxford-IIIT Pet	7,349 images; 37 breeds, species and breed labels, trimaps and head boxes	Train, validation and test	CC BY-SA 4.0; official Oxford source
Team-collected	401 retained photographs, 503 body boxes and masks	Previously evaluated supplemental external benchmark	Private; consent and provenance reported by the team

Oxford-IIIT Pet: https://www.robots.ox.ac.uk/~vgg/data/pets/ (Parkhi et al., 2012). The full dataset is used; no subset was selected before cleaning.

2.2 Cleaning

Check	Method	Removed
Unreadable images or small images	Pillow open/verify; minimum side 32 pixels	0
Exact duplicates	MD5 of file bytes	0
Near duplicates	pHash Hamming distance ≤ 4; union-find grouping	0; grouped
Empty trimaps	No usable foreground mask or body box	14

Retained: 7,335 images, including 2,366 cats and 4,969 dogs across 37 breeds. Median image size is 500 × 375 pixels. Near-duplicate audit: 45 pairs, 41 merged groups covering 84 images; largest group size 3.

2.3 Split and duplicate control

Grouping key: near-duplicate group (pHash Hamming distance ≤ 4, merged transitively), so every image and its near-copies form one group; 7,292 groups in total.

Duplicate control: near-identical photographs are grouped before splitting. The audit found zero duplicate groups and zero MD5 matches crossing splits. Oxford-IIIT Pet has no individual-animal identifiers, so different photographs of the same pet may still cross splits. Similarity filtering below is a sensitivity analysis; it cannot measure or bound a causal identity-leakage effect.

We used a 70/15/15 split by image count, stratified by breed at group level (data.py, group_stratified_split), with seed 42.

Split	Groups	Images	Cats	Dogs	Smallest breed
Train	5,091	5,123	1,652	3,471	128
Validation	1,103	1,106	357	749	28
Test	1,098	1,106	357	749	28

Audit: zero duplicate groups and zero MD5 matches cross splits. Group stratification controls the documented duplicate relation; it does not establish independence of individual animals or pretraining images.

2.4 Image similarity sensitivity analysis

We paired each test image with its nearest same-breed training image using ImageNet ResNet-50 features (median similarity 0.80; 99th percentile 0.955; maximum 0.98). We then recomputed accuracy after filtering close matches. This sensitivity analysis changes the evaluated population; similarity does not verify animal identity.

Test images kept	Removed (close match)	n	Fine-tuned accuracy (95% CI)	Scratch accuracy (95% CI)
All	0	1,106	0.999 (0.995–1.000)	0.906 (0.887–0.922)
Similarity < 0.95	22 (2.0%)	1,084	0.999 (0.995–1.000)	0.905 (0.886–0.921)
Similarity < 0.90	133 (12.0%)	973	0.999 (0.994–1.000)	0.902 (0.882–0.919)
Similarity < 0.85	308 (27.8%)	798	0.999 (0.993–1.000)	0.892 (0.869–0.912)
Similarity < 0.80	562 (50.8%)	544	0.998 (0.990–1.000)	0.884 (0.855–0.908)
Similarity < 0.75	790 (71.4%)	316	0.997 (0.982–0.999)	0.851 (0.808–0.886)

Test split; 224 × 224 inputs; argmax; Wilson intervals. Fine-tuned accuracy is 1105/1106 = 0.999096 on the full test set and 315/316 = 0.996835 at the strictest filter. The difference is 0.226 percentage points across different subsets, not a causal estimate of leakage. Individual-animal independence remains unverified.

Interpretation and limits

The selected model remains accurate on the retained subsets. However, filtering changes test difficulty, breed mix and class composition. Similarity is a proxy and does not confirm animal identity. The experiment cannot prove absence of identity leakage, quantify its causal effect or provide an upper bound on leakage.

These subsets were examined after the final test results were available. They are diagnostic analyses, not additional independent final tests and not a basis for selecting checkpoints or thresholds.

The nearest-neighbour feature extractor uses ImageNet weights. Possible overlap between pretraining images and the Oxford-IIIT Pet dataset remains unexcluded.

Generalisation

Domain bias: web and pet-site photos, mostly one large, centred, well-lit animal (1,105 of 1,106 test boxes are ‘large’). Small, partly visible and multiple animals are under-represented, which is where shelter photos differ. ImageNet and Oxford-IIIT Pet were both collected from the web, so some overlap between pretraining images and the test split cannot be excluded.

Annotation quality: zero species errors in 200 audited images (approximate 95% upper bound 1.5%); four of 300 derived body boxes were loose. Missing animal identities prevent an identity-disjointness claim. Similarity sensitivity does not quantify causal leakage.

Licence: Oxford-IIIT Pet is CC BY-SA 4.0; images are downloaded from the official source and not redistributed.

The benchmark is dominated by well-lit web photographs of one large pet. Shelter photographs may contain multiple animals, unusual poses, occlusion, small objects and clutter. Strong benchmark scores do not establish reliable shelter-domain performance.

2.5 Preprocessing augmentation and dataset distribution

Input resolution is 224 × 224 pixels, with ImageNet mean and standard deviation for normalisation and batch size 64. The training loader uses torch.Generator with seed 42.

RandomResizedCrop to 224 px, scale 0.5–1.0: adoption photos are framed loosely and, in Phase 2, the classifier receives detector crops; random crops simulate both.

Random horizontal flip (p = 0.5): a mirrored cat or dog is still the same species, and pets face both ways.

ColorJitter (brightness, contrast and saturation 0.2; hue 0.02): shelter photos vary in lighting and camera; the hue change is kept small so that coat colour, which is informative, is preserved.

Validation and test: resize to 255 px and centre crop 224 px, without augmentation.

Class imbalance (about two dogs per cat) is handled by breed stratification and by macro-F1 as the primary metric; no class weights or resampling.

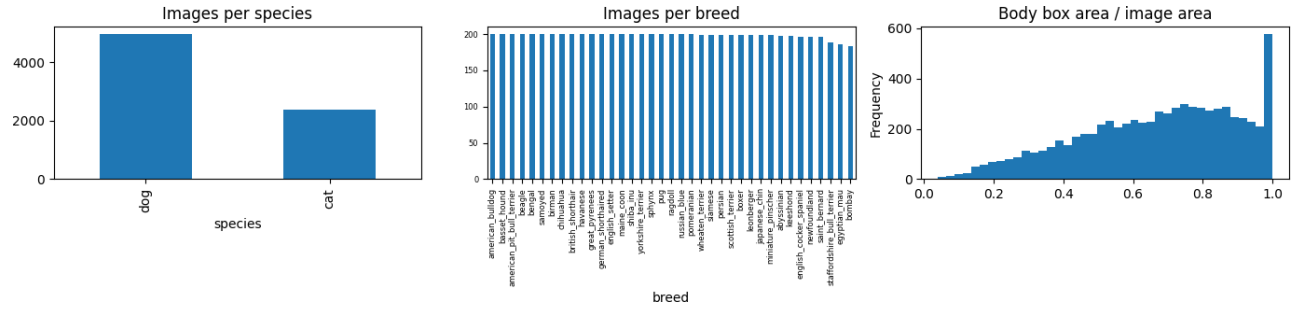


Figure 1. Cleaned dataset distribution: species, breeds and body-box area fraction. All 7,335 retained images; original resolution; body boxes derived from trimaps; no decision threshold.

Data loader and split roles

The training loader is shuffled with a seeded generator. Training uses augmentation and dropout; validation and test use deterministic transforms. Their losses and accuracies therefore need not follow the same trajectory. Normalisation uses ImageNet mean and standard deviation.

The split metadata is restored under runs/data/metadata.csv. The classification code uses those roles; it does not use the 401 private team photographs to train the Phase 1 models.

2.6 Annotation audits and private team data

Audited sample size	Errors found	Estimated label-error rate	Upper bound (95% confidence)
200 images (species label)	0	0.0%	1.5% (one-sided, rule of three)
Same 200 (breed label)	4 doubtful	2.0%	5.0% (Wilson interval 0.8–5.0%)
300 images (derived body box)	4 loose boxes	1.3%	3.4% (Wilson interval 0.5–3.4%)

We inspected contact sheets sampled using the project seed and confirmed the audit findings as a team. The audit records retain reviewer and confirmation fields. Zero observed species errors does not prove that every label is correct; the rule-of-three confidence bound is approximate.

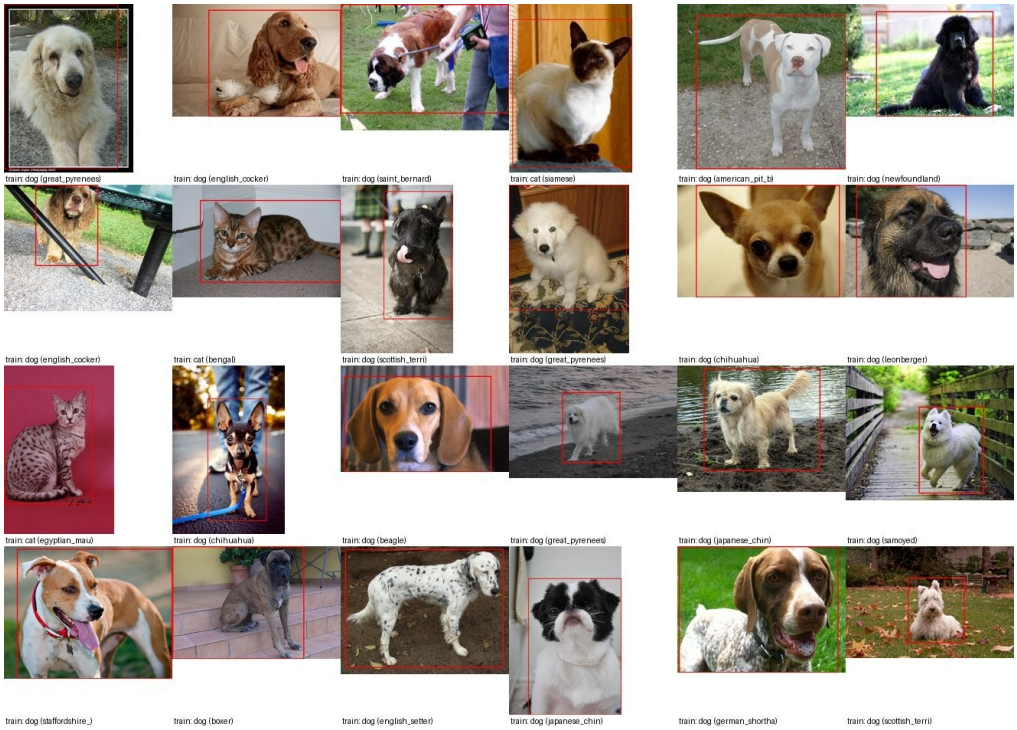


Figure 2. Examples from the body-box review. Training split; original resolution; no threshold. Boxes are derived from trimap foreground plus border rather than the dataset head-box annotations.

The 401-photo dataset is private to the four team members and excluded from the public repository and release packages. Our collection records document consent and provenance, metadata removal and the exclusion of 13 suspect photographs. CVAT review corrected eight species labels. This dataset has already been evaluated.

We retained 401 of 414 collected photographs: 178 cat, 221 dog and 2 mixed-species images. The dataset contains 401 masks and label files, with 503 body boxes matching the recorded animal counts. We describe the collection as team-collected and team-annotated.

3 Classification M2 and M3

3.1 Models and training protocol

Models are selected by validation macro-F1, using the first epoch reaching the maximum. The held-out test split evaluates the retained models; it is not used for selection. Common recipe: AdamW, weight decay 1e-4, cosine schedule, batch 64, mixed precision, cross-entropy and seed 42. Temperature is fitted on validation only. Scratch uses five convolutional blocks with BatchNorm/ReLU/max-pooling, global average pooling, dropout 0.3 and a linear layer (2,354,402 parameters). The frozen model trains only a new 2048 → 2 head (4,098 parameters) with BatchNorm statistics frozen; full fine-tuning trains all 23.5 M parameters from the ImageNet weights with a ten times smaller learning rate. All runs are in W&B group full-rerun-2026-10-18.

Strategy	Backbone	Trainable parameters	Epochs	LR	Best epoch
Scratch	Custom CNN	2,354,402	25	0.001	25
Frozen	ResNet-50	4,098	8	0.001	6
Fine-tuned	ResNet-50	23,512,130	10	0.0001	6

Frozen training updates the new 2048-to-2 head and keeps BatchNorm statistics frozen. Fine-tuning updates all ResNet parameters. First maximum validation macro-F1 selects the retained checkpoint. Different architectures, budgets and learning rates make this an application comparison, not an isolated causal ablation.

Model	Best validation accuracy	Best validation macro-F1	Original W&B run
Scratch	0.9276672694	0.9175159319	gh7ayx6h
Frozen	0.9981916817	0.9979288390	xsz5afr9
Fine-tuned	1.0000000000	1.0000000000	8nvxbsn2

Final held-out test results

Model	Correct / n	Accuracy	Macro-F1	95% Wilson CI
Scratch	1002 / 1106	0.9059674503	0.8922996255	0.887–0.922
Frozen	1105 / 1106	0.9990958409	0.9989651820	0.995–1.000
Fine-tuned	1105 / 1106	0.9990958409	0.9989651820	0.995–1.000

Test n = 1,106; 224 × 224; argmax; no threshold. Fine-tuned best epoch 6: validation accuracy 100%; last epoch 10: 99.9096%. These validation values are not test results.

Selected model: fully fine-tuned ResNet-50, chosen by best validation macro-F1 (1.000 versus 0.998 for frozen). It meets both Phase 1 targets; scratch misses macro-F1 (0.892). Exact McNemar p = 1.0 detects no difference between transfer models on this test set; it does not establish equivalence. Both differ from scratch ($p < 1e-28$). Frozen is a lower-cost alternative.

3.2 Learning curves

We repeated the three training configurations with full metric logging, using the same configuration, seed and data order. Shared training-loss, validation-accuracy and macro-F1 histories agree within 1e-12; runtimes differ. Training accuracy and validation loss are rerun measurements, not recovered original logs. Original submitted checkpoints were retained.

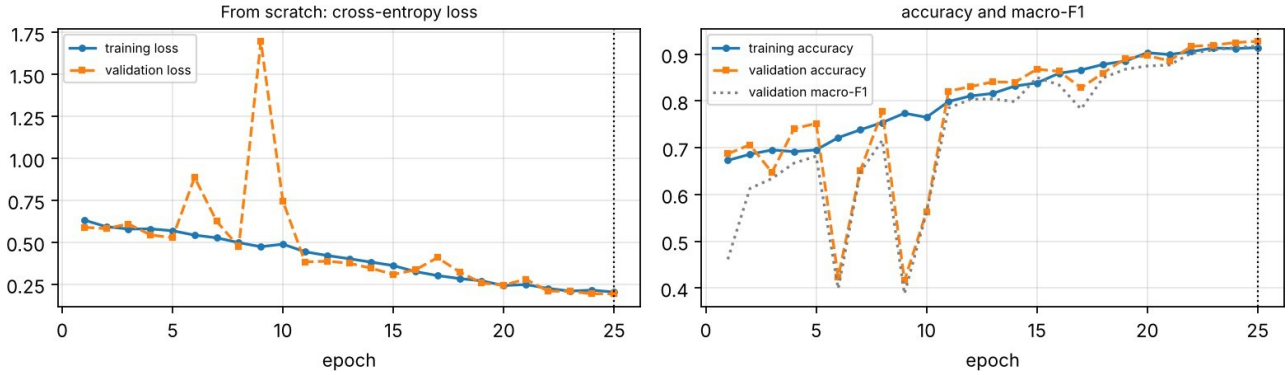


Figure 3a. Scratch CNN: training/validation loss and accuracy, with validation macro-F1. Full-logging rerun bvns5vir; validation n = 1,106; 224 × 224 inputs; argmax.

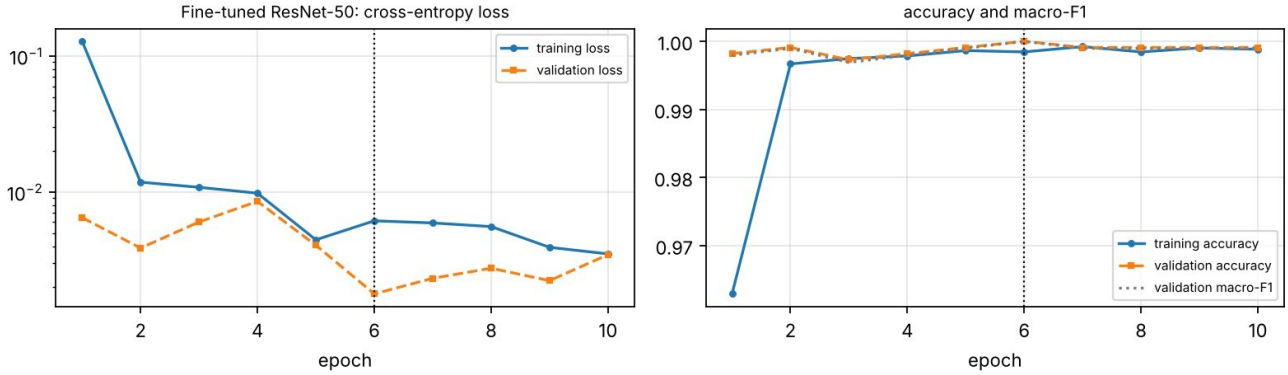


Figure 3b. Fine-tuned ResNet-50: same metrics; log loss axis; full-logging rerun b3t0emrs. The dotted line identifies the retained checkpoint. Training metrics average augmented batches in training mode.

Scratch appears under-trained: at epoch 25 training accuracy is approximately 0.914 and validation accuracy 0.928; training loss is approximately 0.204 and validation loss 0.193. Augmentation/dropout contribute to the gap. Validation spikes may reflect evaluation-mode BatchNorm instability; this is an interpretation, not a measured causal effect.

Fine-tuned validation loss reaches approximately 0.0018 at epoch 6 and rises to 0.0035 by epoch 10, consistent with mild overfitting after the selected checkpoint. Frozen validation loss decreases from approximately 0.063 to 0.013; its full curve is recorded in W&B run 4ixtdkiq.

3.3 Per class results and calibration

Class	Precision	Recall	F1	Support
Cat	1.000000	0.997199	0.998597	357
Dog	0.998667	1.000000	0.999333	749
Macro average	0.999333	0.998599	0.998965	1106

Selected fine-tuned model; test split; 224 × 224; argmax. Binary top-2 accuracy is trivially 1.0 and is not informative.

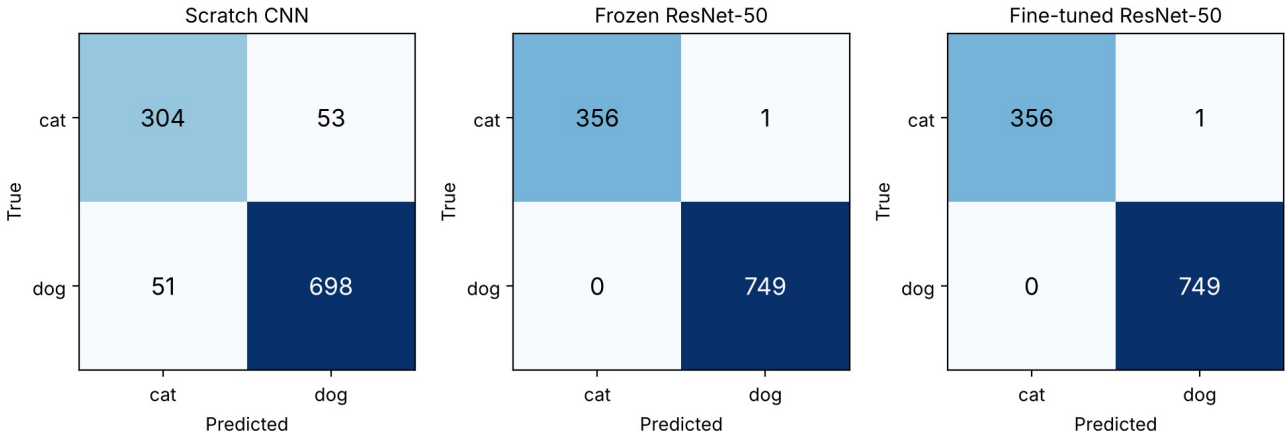


Figure 4. Test confusion matrices. Rows are true labels and columns predicted labels, in cat/dog order. Scratch [[304,53],[51,698]]; frozen and fine-tuned [[356,1],[0,749]]. Test n = 1,106; 224 × 224; argmax; no threshold.

Model	ECE raw	ECE calibrated	Saved temperature
Scratch	0.0135853811	0.0247622819	0.8230744600
Frozen	0.0109669426	0.0012126884	0.4895465076
Fine-tuned	0.0014543669	0.0009041591	0.05 lower bound

ECE uses 15 equal-width confidence bins. Temperature is fitted only on validation. Scaling worsens scratch ECE and lowers transfer-model ECE, but the small differences do not establish statistical significance or shelter-domain calibration. Fine-tuned temperature reaches the optimization bound. The provisional 0.90 referral threshold is not certified for deployment.

Equal frozen and fine-tuned accuracy and confusion matrices do not imply identical predictions. Checkpoint inference on 28 October identified two image-level disagreements. Their single errors were a Sphynx cat and a Bombay cat, respectively.

3.4 Reliability and error analysis

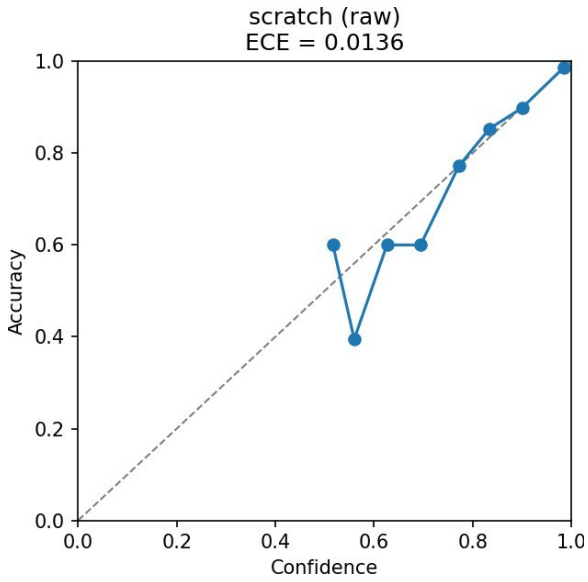


Figure 5a. Scratch reliability before scaling. Test n = 1,106; 15 confidence bins; 224 × 224; argmax.

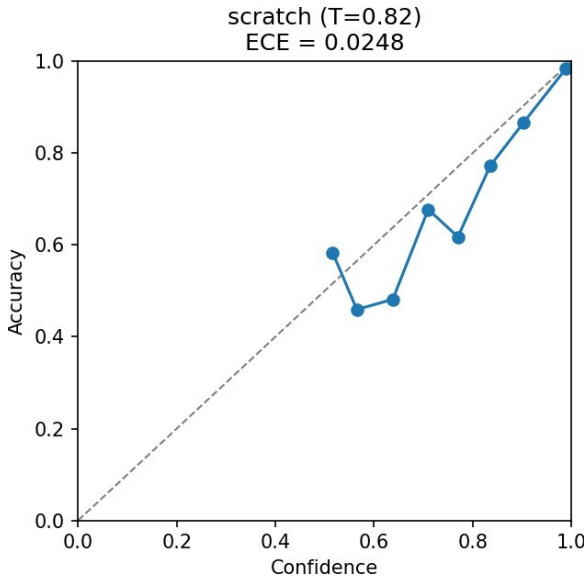


Figure 5b. Scratch reliability after scaling, T ≈ 0.823. Same evaluation protocol; calibration worsens ECE.

Reliability diagrams for both transfer models and the other raw/calibrated comparisons are retained in the classification figures directory. Near-perfect validation separation limits the interpretation of confidence calibration. Lower aggregate ECE does not guarantee calibrated confidence for individual shelter photographs.

Most confused direction for the selected model is cat to dog. Scratch makes 53 cat-to-dog and 51 dog-to-cat mistakes. Coat texture, low contrast and curled-up pose are hypotheses suggested by gallery inspection; controlled experiments were not performed to establish those causes.

3.5 Error gallery and improvement plan

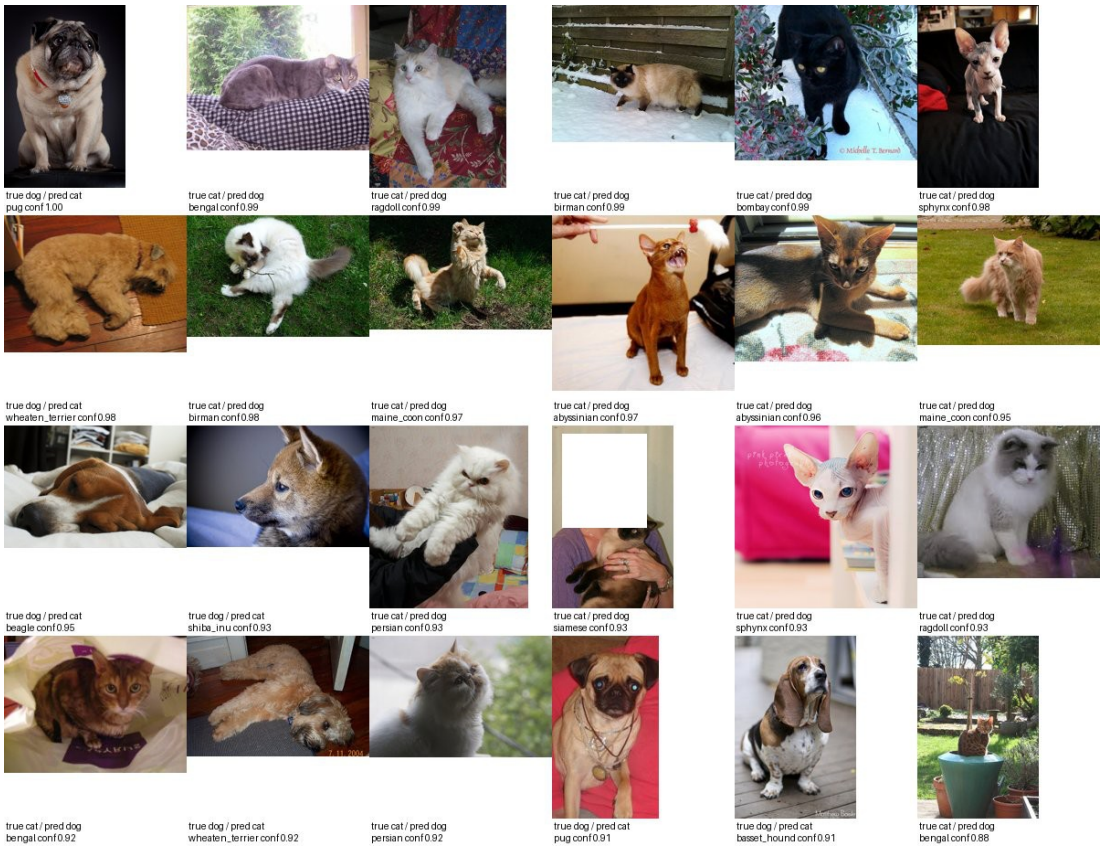


Figure 6. Most confident scratch-CNN errors. Test split; 224 × 224; argmax; no threshold. Visible face redacted for display only; model inputs and metrics unchanged.

Finding	Examples	Interpretation
Long-haired and colour-point cats	Ragdoll 5; Birman 5; Persian 4; Maine Coon 4 errors	Fur/shape cues may be hard for the small CNN
Hairless and black cats	Scratch: Bombay 7/28; Sphynx 6/30	Appearance and contrast may contribute
Curled-up dogs	Pug 5; wheaten terrier 5; beagle 3; basset hound 3	Pose hides shape cues
Label noise	0 species errors in the 200-image audit	Not excluded on individual images

Counts are breed-level test errors. Of the 24 displayed scratch errors, 17 are cats predicted as dogs. These are descriptive observations, not quantified causal effects.

3.6 Data improvement and limitations

Finding	Development action	Evaluation rule
Cat appearance errors and 2:1 dog imbalance	Collect more black, hairless and colour-point cats; assess species rebalancing	Choose changes on training/validation only
Small, occluded and unusual-pose animals are rare	Collect harder development photographs, including puppies and multiple animals	Use a new independent final test after decisions are fixed
Temperature hits its lower bound	Compare calibration approaches on a harder separate calibration set	Keep calibration and final test roles separate
Four loose body boxes	Review trimap-derived boxes and add quality checks	Evaluate detector improvements in Phase 2

The 401-photo collection is a previously evaluated supplemental external benchmark. Its integrated-pipeline metrics concern Phase 2 and do not replace Phase 1 whole-image classification results. Further development will require a new independent final test after model and threshold choices are fixed.

Limits of the evidence

Verified animal identities are unavailable, and possible ImageNet/Oxford overlap cannot be ruled out. Duplicate grouping is necessary but is not a guarantee against every form of leakage.

There are no independent multi-seed runs establishing training variance. Different architectures, learning rates and training budgets confound causal comparison.

High benchmark accuracy and small ECE are not evidence that shelter photographs are safe for automatic publication. Staff retain publishing authority.

Our collection records document team-photo consent and provenance; these have not undergone external certification.

The evidence comprises original training results, full-logging reruns, checkpoint inference and saved-output verification. These operations address different aspects of reproducibility; they do not provide independent multi-seed estimates.

4 Reproducibility

Item	Answer
Repository and version	edcc29f2e035b329e577cd04bf6092dc1f4ae04c, checked on 29 October 2026. This is the published source snapshot, not an established original-training commit.
Configuration	smart-pet-vision/configs/config.yaml; seed 42 for Python, NumPy, PyTorch and data loading. requirements.txt pins the training dependencies.
Checkpoints	Release phase1-v1.0: Phase1_Checkpoints.zip contains scratch.pt, frozen.pt and finetune.pt. Checkpoints use the first maximum validation macro-F1.
Saved results	Phase1_Results_v2.zip contains data/, classification/ and verification_2026-10-28/ inside "مجلد جديد". These directories correspond to smart-pet-vision/runs/.
Verification protocol	Saved-output verification recalculates metrics from stored predictions. Checkpoint inference evaluates existing weights. Retraining optimizes parameters and produces new records; notebooks 01 and 02 implement this workflow.
Data	Oxford images are downloaded from the official VGG source. Phase 1 training does not require the private team dataset. Public result archives contain metadata and results, not original Oxford image files.
Training reproduction	The training workflow consists of notebooks 01 and 02 in a separate environment. The archive and saved-output checks did not involve retraining. Separately, clean Colab T4 retraining was completed on 29 October 2026, as documented below.
Historical team record	Our team record identifies Yasser Alhajlah's clean Colab T4 reproduction on 23 October 2026, with matching reported metrics. This is distinct from the 28 October checkpoint verification.
Additional verification	28 October: float32 MPS inference with all three weights on 2,212 MD5-matching validation/test JPEGs. 29 October: public archive SHA-256 checks and metric recomputation from stored outputs, without retraining.
Availability limitations	The public source and results archive do not include MANIFEST_SHA256.txt, RELEASE_VERSION.json, Phase1_Source.bundle, executed notebooks or runs/supplement/. The public PDF build helper lacks its required base input, so a public clean-checkout PDF rebuild is not established.

Public release links

https://github.com/Yasser-20-alhajlah/smart-pet-vision/releases/tag/phase1-v1.0

The checkpoint and results archives are publicly available in release phase1-v1.0. Both were downloaded and verified on 29 October 2026.

Checkpoints: https://github.com/Yasser-20-alhajlah/smart-pet-vision/releases/download/phase1-v1.0/Phase1_Checkpoints.zip
Results: https://github.com/Yasser-20-alhajlah/smart-pet-vision/releases/download/phase1-v1.0/Phase1_Results_v2.zip

Clean retraining verification — 29 October 2026

All three classifiers were retrained and evaluated in a separate Google Colab T4 working directory using seed 42. The scratch, frozen and fine-tuned runs completed 25, 8 and 10 epochs, respectively. Checkpoints were selected using validation macro-F1.

Across all recorded epochs, the five logged learning-curve metrics matched the previous full-logging reference runs. The resulting test accuracy and macro-F1 also matched the reference results. This verifies same-seed reproducibility; it does not estimate variability across independent seeds.

The new outputs were saved separately from the original submission evidence.

Model	Run ID	Test accuracy	Test macro-F1
Scratch	`9fay48ne`	0.905967	0.892300
Frozen	`qlzja26v`	0.999096	0.998965
Fine-tuned	`2ptjc52f`	0.999096	0.998965

5 Obstacles and conclusions

Obstacle	Action and result	Status
Head boxes only	Derived full-body boxes from trimaps; reviewed 300; four loose boxes recorded	Addressed, residual annotation error
14 empty trimaps	Removed during cleaning; retained population explicitly reported	Addressed
Missing individual-animal IDs	Grouped near duplicates; reported similarity sensitivity without claiming a leakage bound	Open limitation
Near-separable calibration data	Reported temperature lower bound and provisional threshold; proposed new calibration set	Open
Privacy and photo provenance	Private team storage, team-reported consent, metadata removal and exclusion of 13 suspect photos	Evidence limits disclosed
Shared-folder overwrite risk	Separate working copy for saved-output verification and for retraining	Controlled by workflow

Conclusion

Full fine-tuning achieves the strongest validation macro-F1 and meets both project targets on the held-out Oxford-IIIT Pet test split. Frozen transfer learning achieves the same test accuracy and macro-F1 at substantially lower training cost. The custom CNN performs less well and falls below the macro-F1 target.

The next phase will address detection and segmentation, harder development photographs and calibration under domain shift. New independent evaluation data will be required after development choices are fixed. Current benchmark results do not establish deployment performance on shelter photographs.

Appendix A Dataset card

Field	Dataset record
Dataset and task	Oxford-IIIT Pet (Parkhi et al., 2012); binary cat/dog whole-image classification. 37 breeds: 12 cat and 25 dog breeds.
Licence and download	CC BY-SA 4.0 as documented by the project. Download from www.robots.ox.ac.uk/~vgg/data/pets/. Images are not redistributed in this submission.
Population	7,349 source images; 14 empty-trimap exclusions; 7,335 retained. 2,366 cats and 4,969 dogs. Median image size 500 × 375.
Cleaning	Pillow verification and minimum side 32 pixels; MD5 exact-duplicate check; pHash Hamming ≤ 4 and union-find groups. 45 pairs, 41 merged groups covering 84 images.
Split	Seed 42; group-aware breed-stratified 70/15/15. Train 5,123 (1,652 cats/3,471 dogs), validation 1,106 (357/749), test 1,106 (357/749). Zero group/MD5 crossings.
Annotations	Body foreground equals trimap foreground plus border. Body box is the tight rectangle around this mask. Border ambiguity matters for later segmentation evaluation.
Audits	200 species labels: zero errors, rule-of-three 95% upper bound approximately 1.5%. Four doubtful breed labels: Wilson CI 0.8–5.0%. Four loose boxes among 300: CI 0.5–3.4%.
Bias and limits	Mostly one large central pet; dogs outnumber cats about 2:1; small, occluded and multiple animals are underrepresented. Missing verified animal IDs and possible pretraining overlap limit independence claims.
Sensitivity study	Nearest same-breed ImageNet-feature matches are filtered diagnostically. Accuracy on retained subsets does not bound causal leakage or establish identity-disjointness.
Intended use	Academic classification comparison and support for a later staff-review workflow. No automatic publishing authorization is established.
Storage	Public metadata, audit exports, model outputs and hashes are separate from Oxford images and private team photographs.

Appendix A Team collected dataset and references

We collected and annotated 401 retained photographs after excluding 13 of 414 received images. The collection records distinguish 104 owner photographs received as screenshots and 297 supplied by team members; 170 of the latter retained camera EXIF before metadata removal. These source categories do not establish who personally photographed every image.

Audit field	Recorded value
Images / masks / labels	401 / 401 / 401
Species	178 cat; 221 dog; 2 mixed
Body boxes	503; counts match n_animals
Readable and aligned	All photos readable; masks match image dimensions
Dimensions and metadata	All photos at most 2048 pixels; collection audit records EXIF removal
Duplicates	No exact duplicates; no reported MD5/pHash match with Oxford
Annotation	Automatic drafts corrected by four teammates in CVAT; eight species-label corrections
Mask share	Median approximately 0.28
Consent	Consent documented in the team collection records; no independent certification

This collection has supported integrated-pipeline evaluation as supplemental Phase 2 evidence. It is not an independent, previously unused final test for subsequent model development.

The team dataset is stored privately under SmartPetVision_TeamData and is excluded from public release packages. The public repository and W&B report contain the instructor-facing project evidence.

References

Phase 2 reference targets, outside the Phase 1 assessment: mAP50 ≥ 0.75, recall ≥ 0.80, mIoU ≥ 0.75, Dice ≥ 0.85, service success ≥ 99%, p95 latency ≤ 2 seconds on T4 and ONNX parity ≤ 0.001. These are team-stated targets, not certified operating performance.

Parkhi, O. M., Vedaldi, A., Zisserman, A., and Jawahar, C. V. (2012). Cats and Dogs. IEEE Conference on Computer Vision and Pattern Recognition. Dataset: https://www.robots.ox.ac.uk/~vgg/data/pets/.

Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. International Conference on Machine Learning. https://proceedings.mlr.press/v70/guo17a.html.

Original W&B runs: scratch gh7ayx6h, frozen xsz5afr9 and fine-tuned 8nvxbsn2. Full-logging reruns: scratch bvns5vir, frozen 4ixtdkiq and fine-tuned b3t0emrs. Checkpoint verification exports are included in the public results archive.